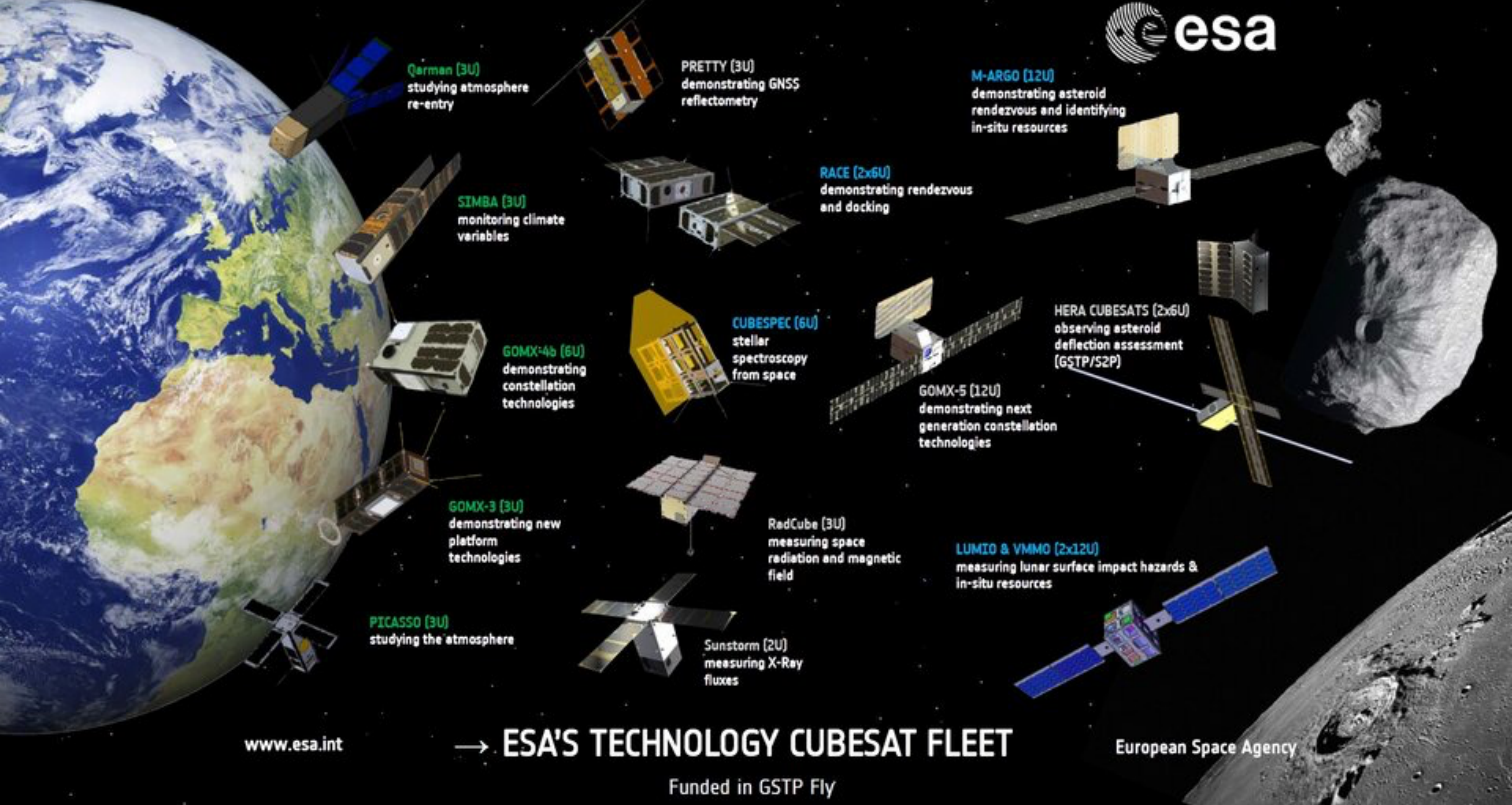


Úvod do FPGA

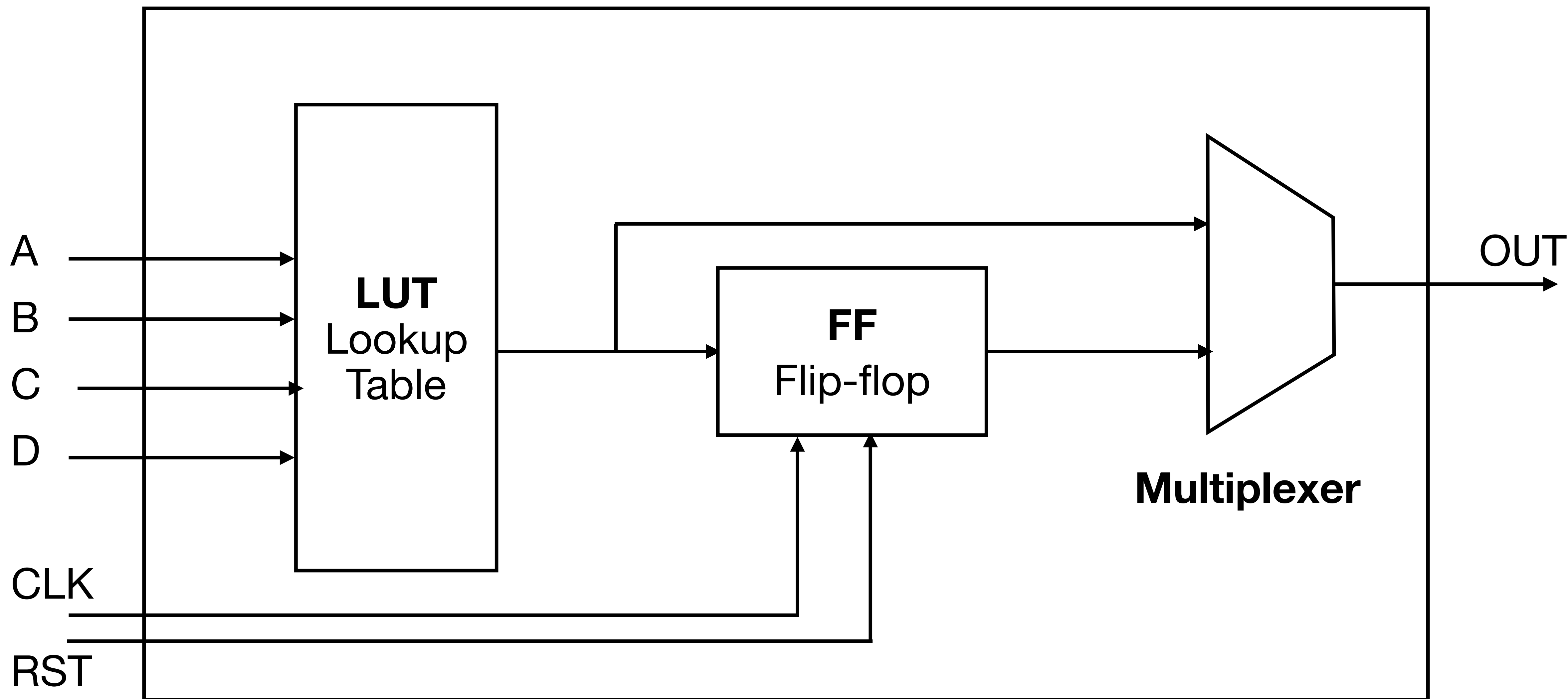
Čo je FPGA čip?

(Field Programmable Gate Array)

FPGA je čip schopný **menit hardware za chodu**
a vytvářet libovolný digitální obvod.



CLB
(Configurable Logic Block)

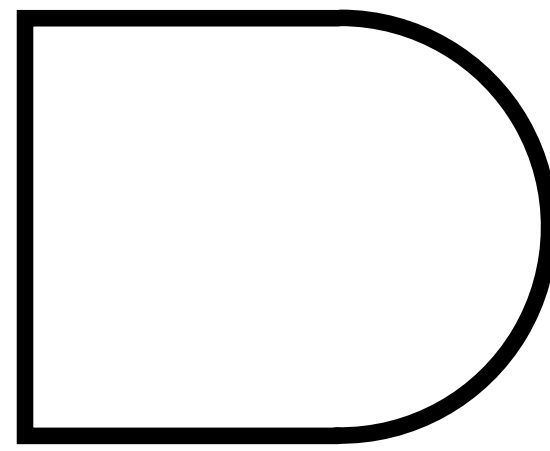


LUT

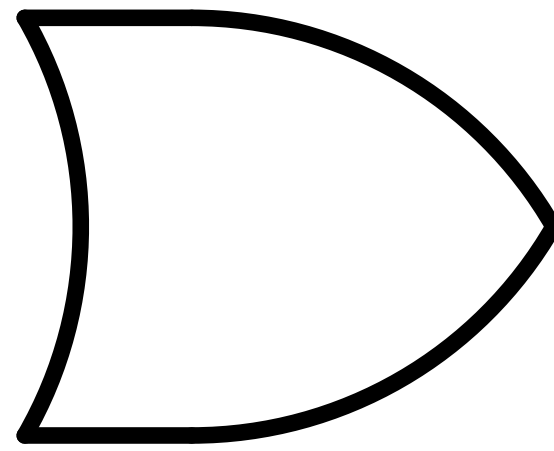
A	B	C	D	OUT
0	0	0	0	1
0	0	0	1	1

⋮

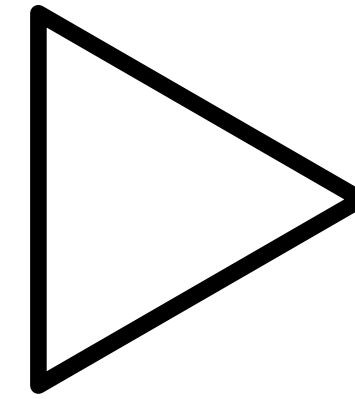
1	1	1	0	1
1	1	1	1	0



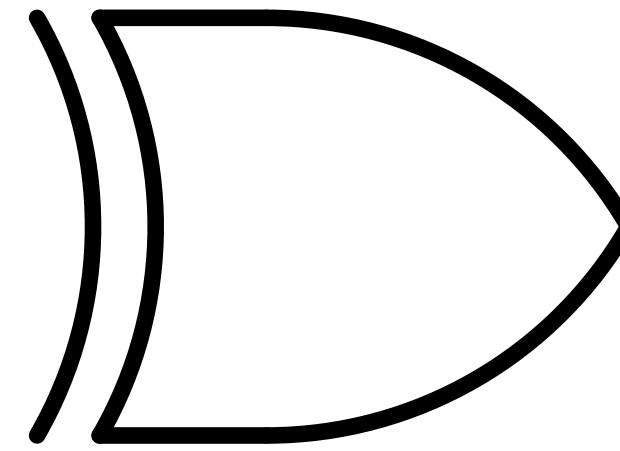
AND



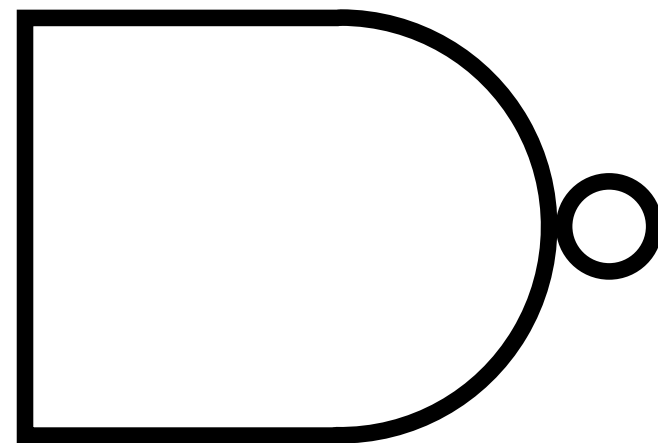
OR



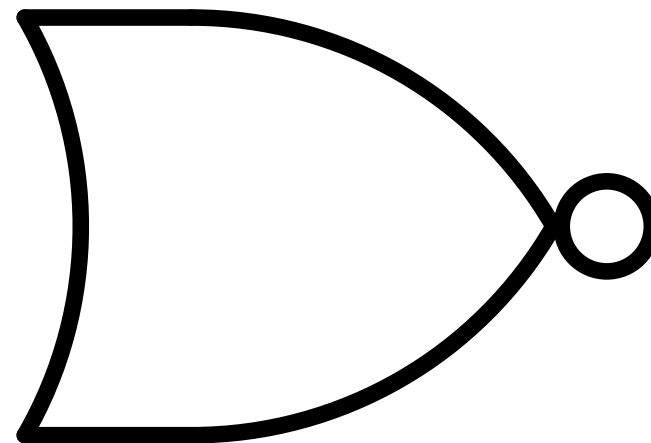
BUFFER



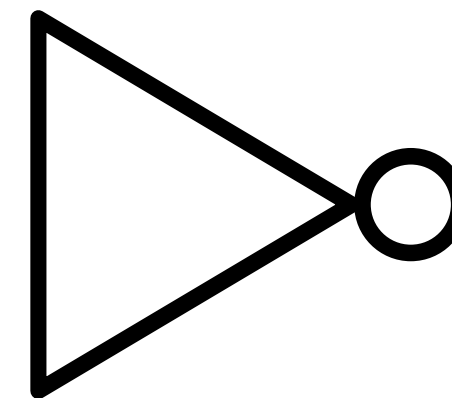
XOR



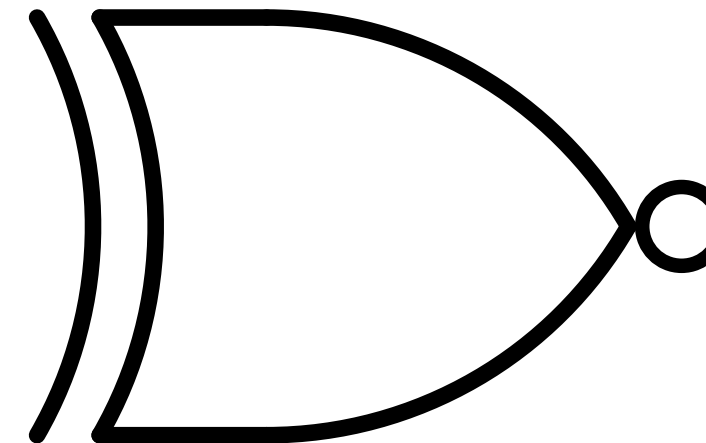
NAND



NOR



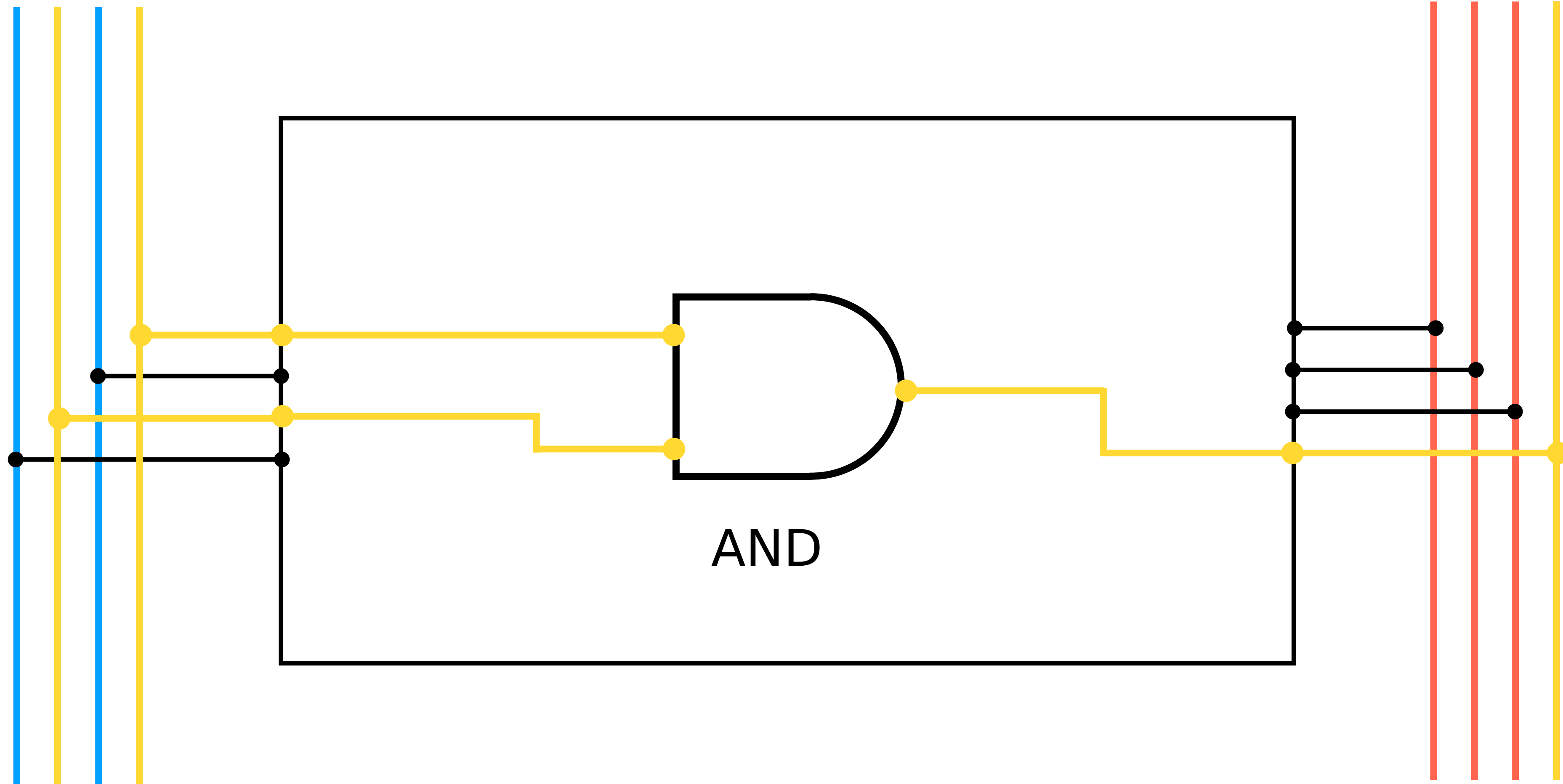
NOT

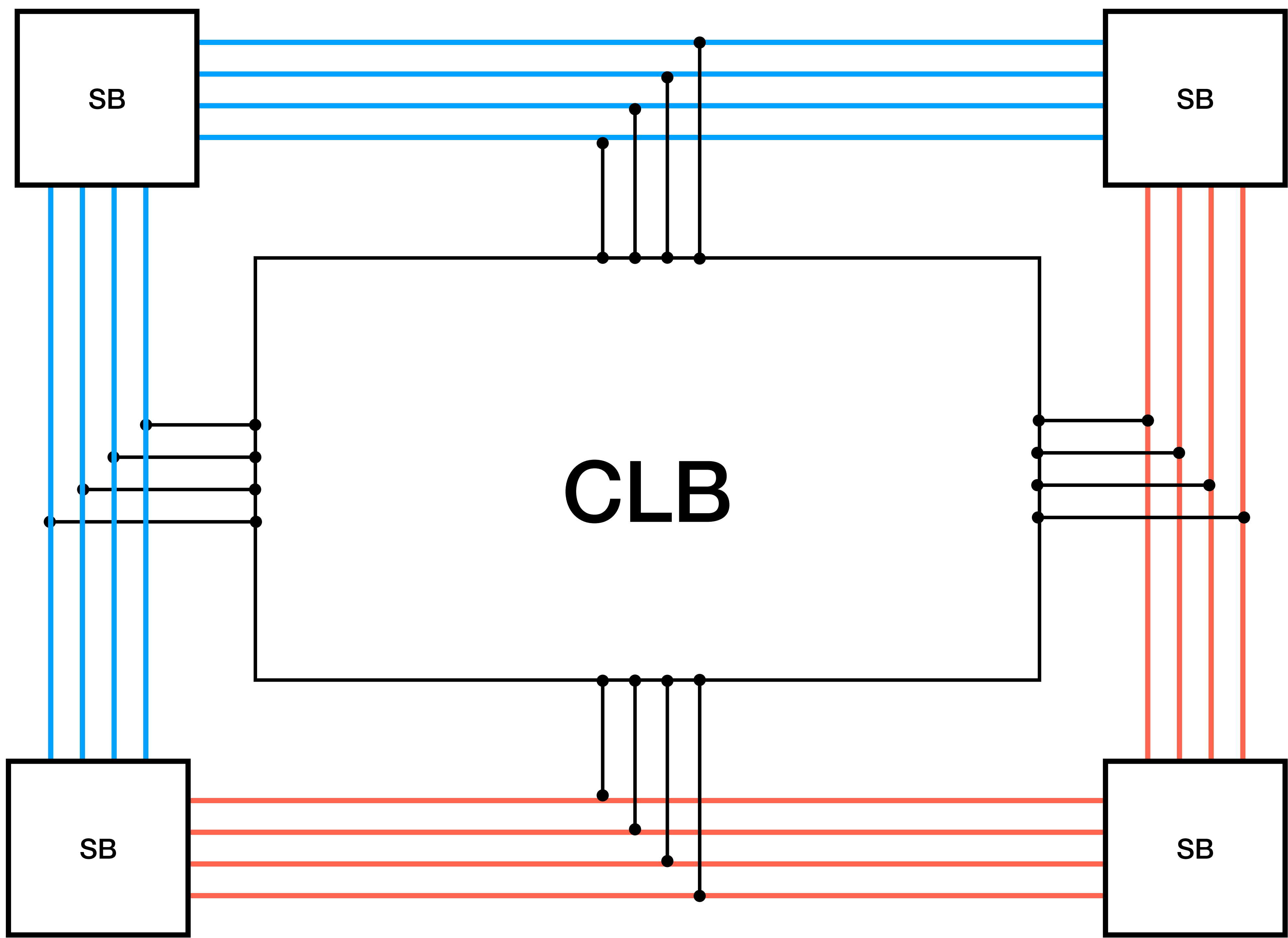


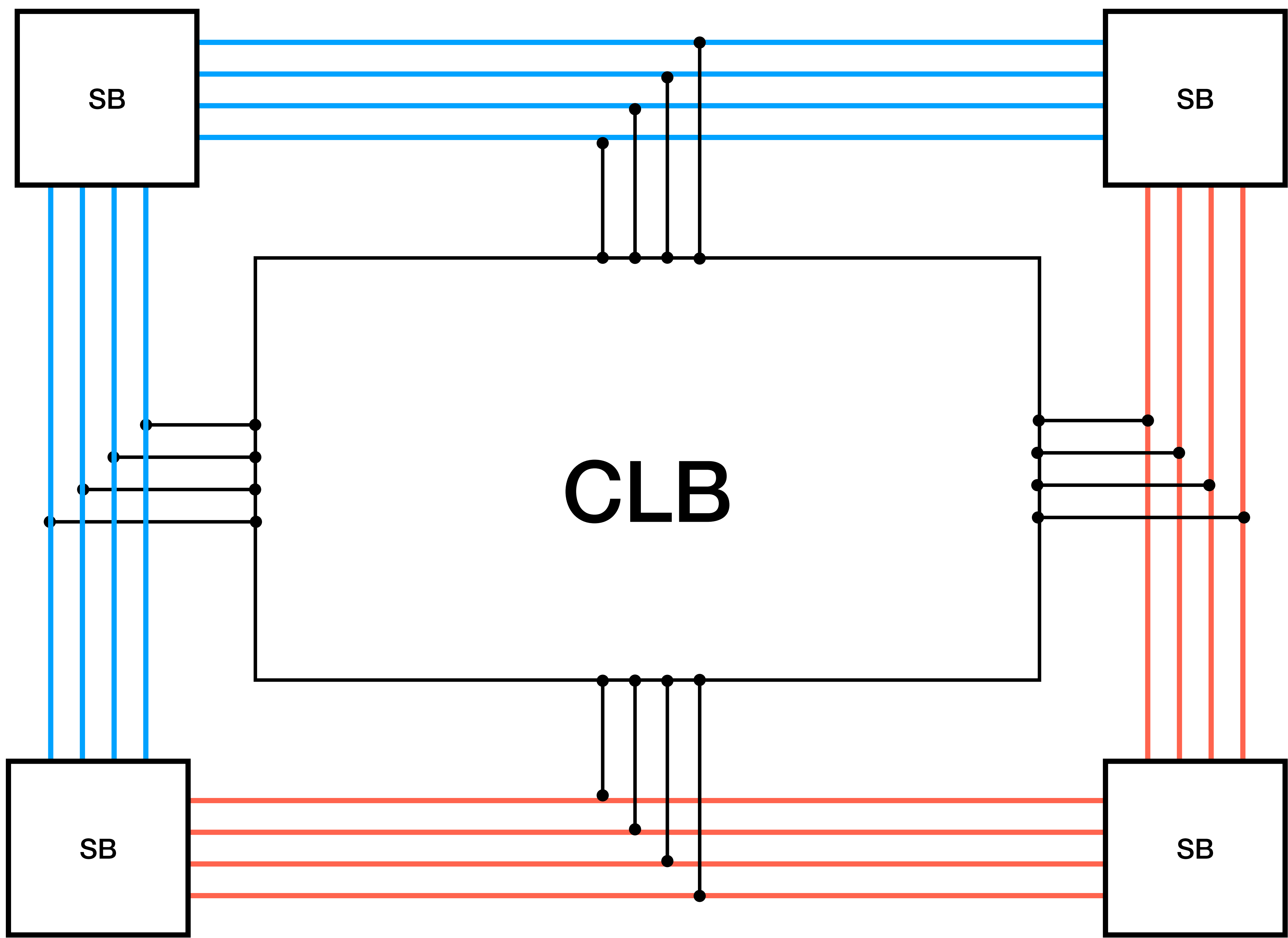
XNOR

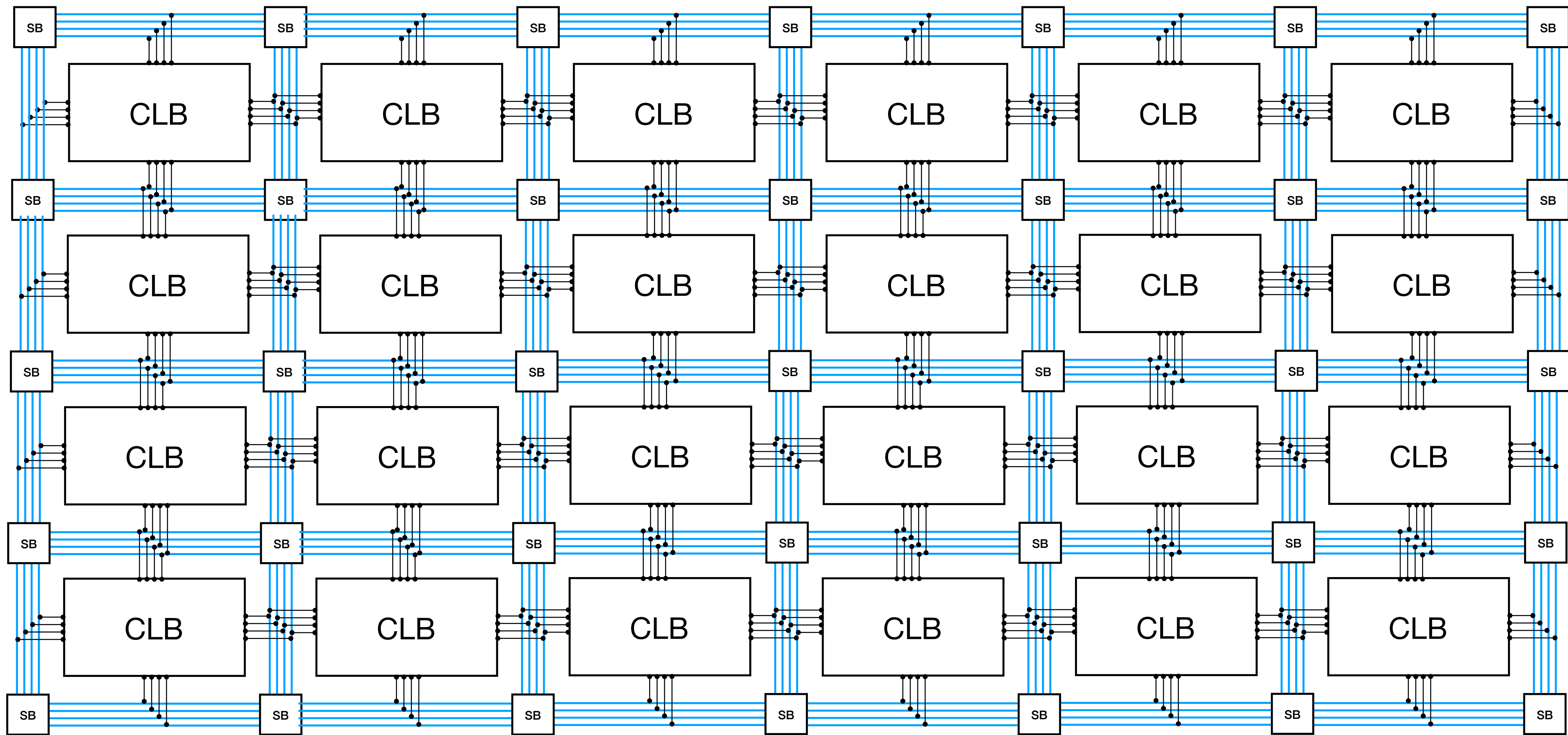
Input

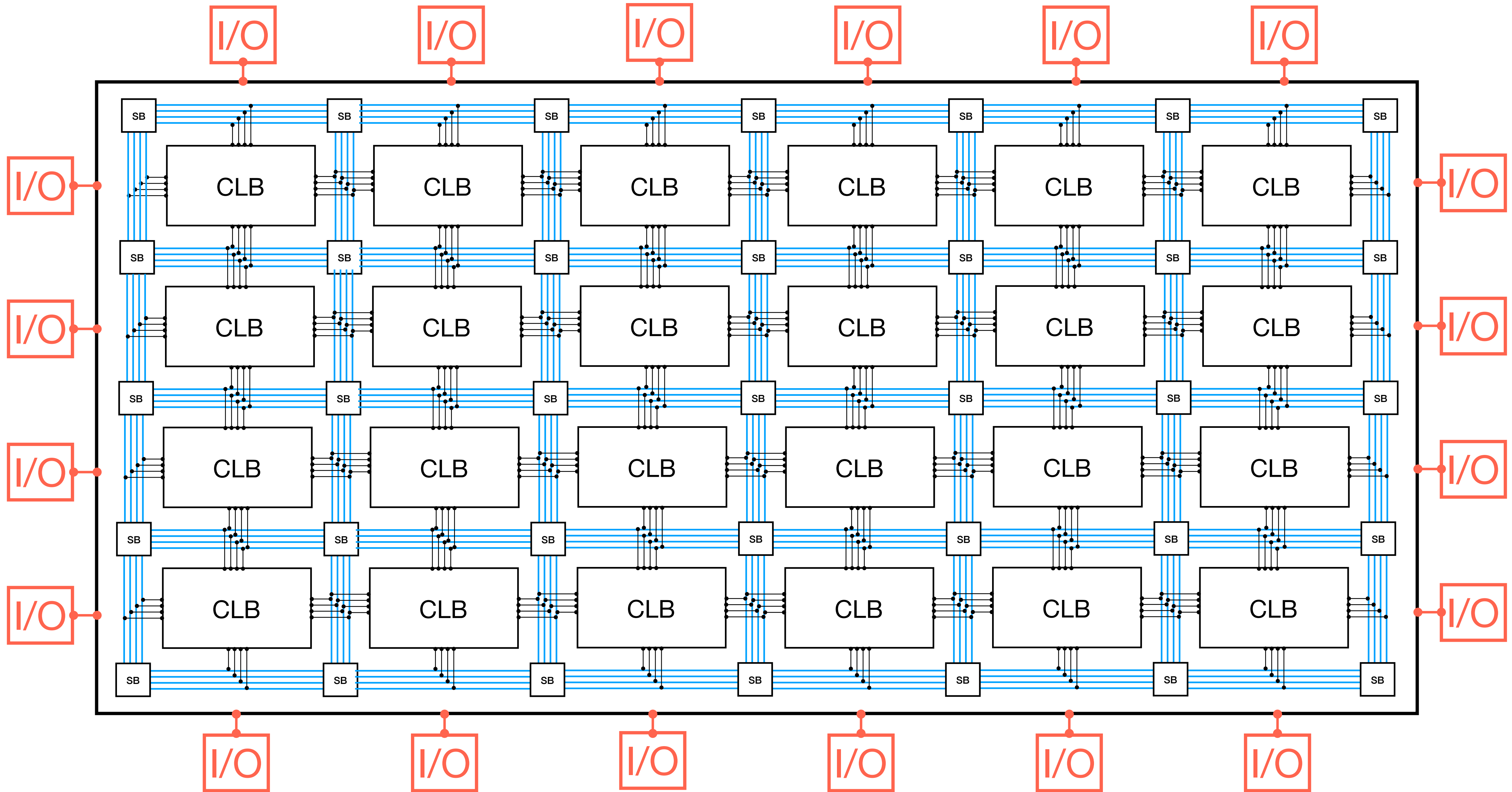
Output

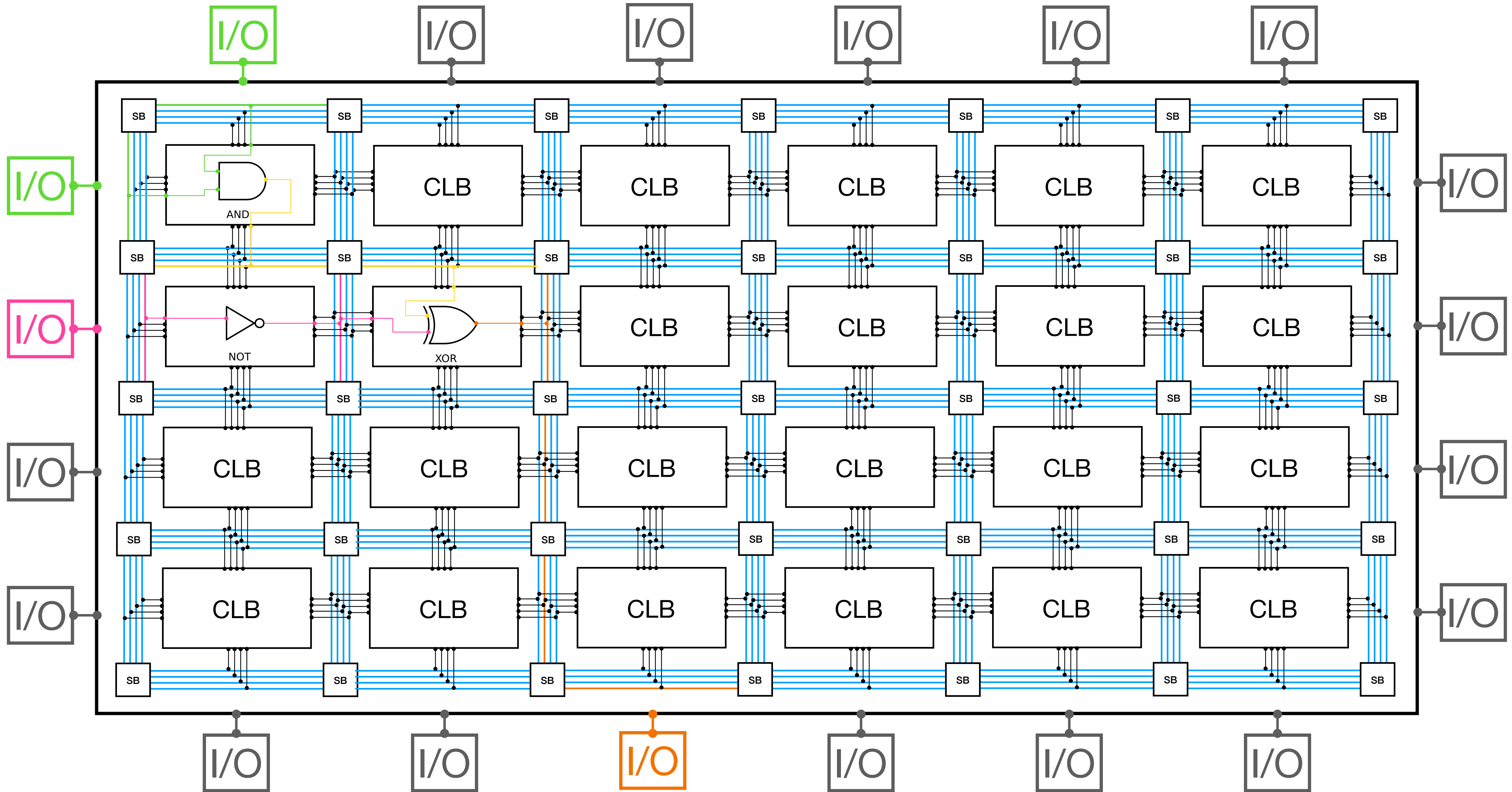




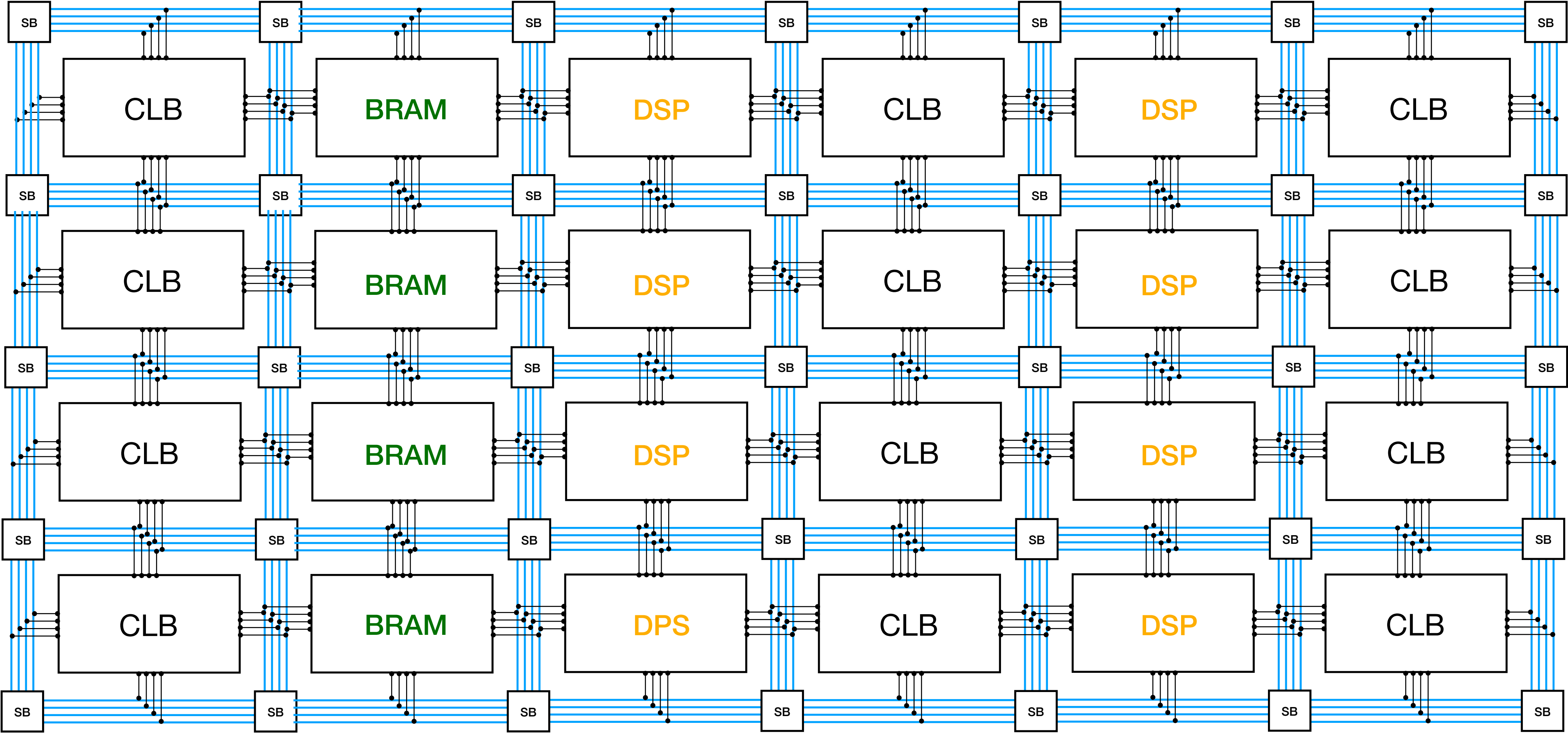




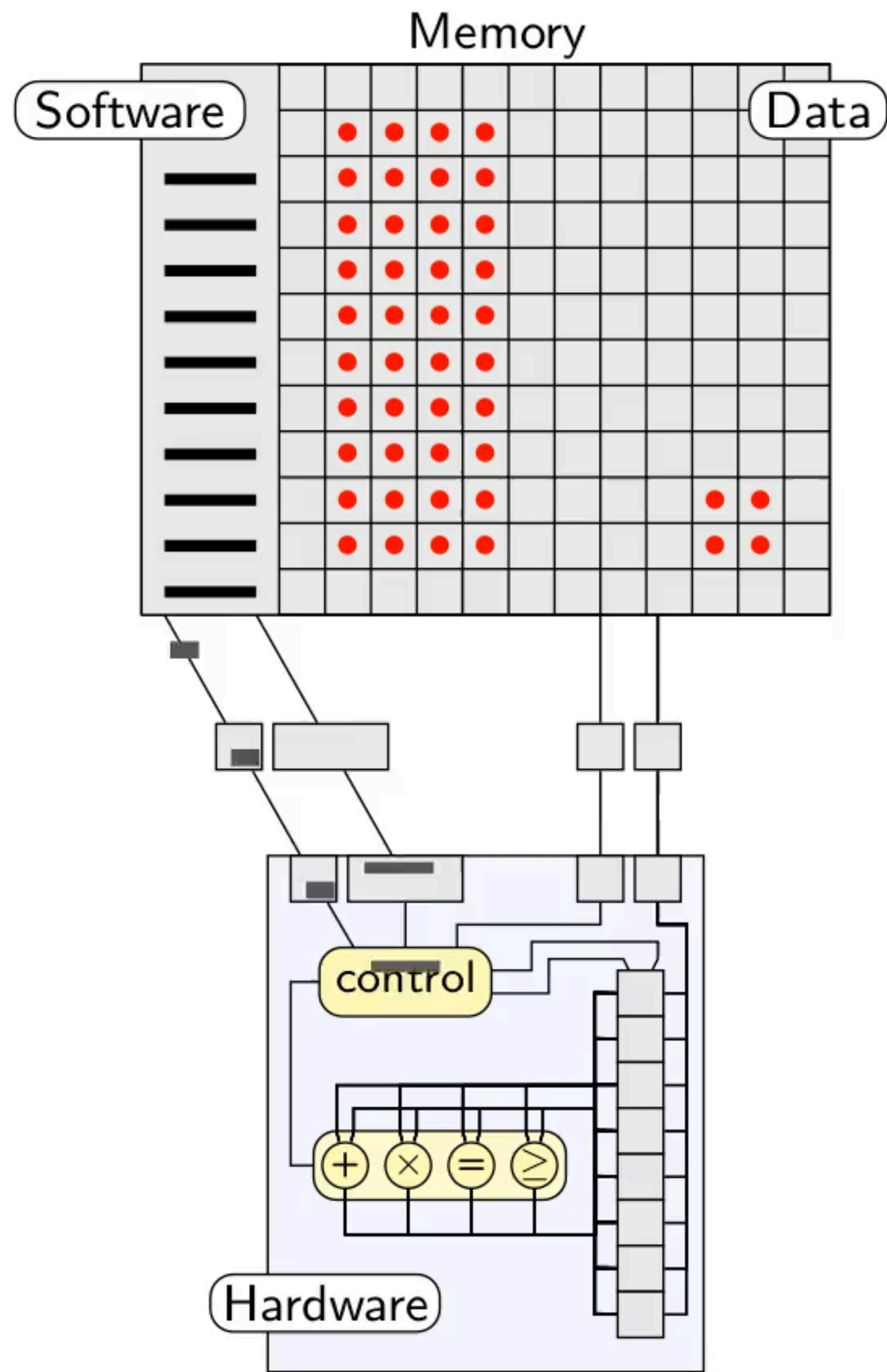




Block RAM Digital Signal Procesor

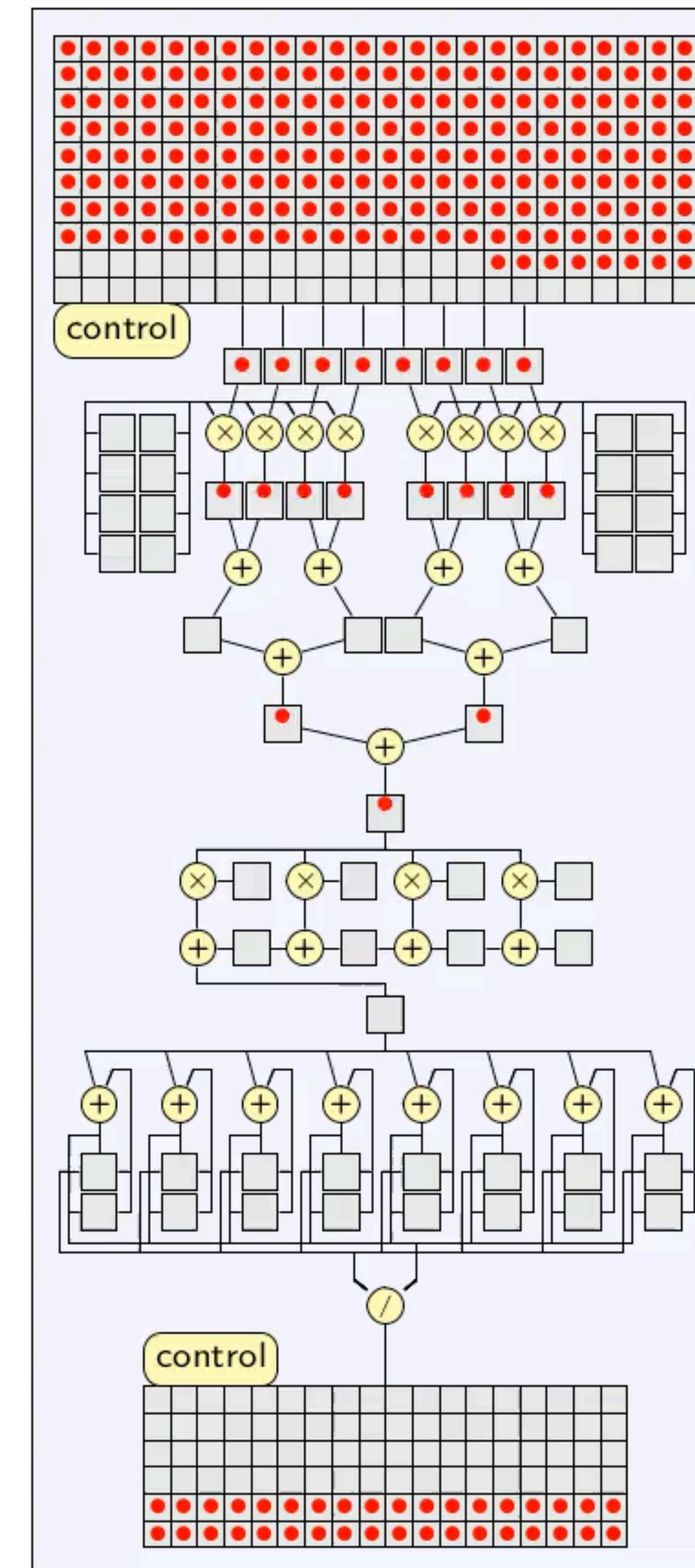


FPGA vs CPU vs GPU



Normal processor

Software controls computation process



FPGA

Computation integrated in hardware

	CPU	GPU	FPGA
Výpočtový model	Sekvenčný	Masívne paralelný	Na mieru
Architektúra	Fixná	Fixná	Flexibilná
Latencia	Stredná (Premenlivá)	Vysoká (Batching)	Nízka (Determinostická)
Spracovanie v reálnom čase *	👎	👎	👍
Periférie (I/O)	Závisí od hostiteľa (PCIe)	Závisí od hostiteľa (PCIe)	Priame HV rozhrania
Energetická efektivita	Nízka	Nízka	Vysoká

Ako sa programujú FPGA čipy?

HDL - (Hardware Definition Language)

VHDL

```
-- Simple Logic Gates  
architecture Behavioral of logic_demo is  
begin  
    and_out <= a and b;    -- Direct AND gate wiring  
    or_out  <= a or b;     -- Direct OR gate wiring  
end architecture;
```

Verilog

```
// Simple Logic Gates  
module logic_demo (  
    input  logic a, b,  
    output logic and_out, or_out  
);  
    assign and_out = a & b;    // Direct AND gate  
    assign or_out  = a | b;    // Direct OR gate  
endmodule
```

HSL - (High-Level Synthesis)

C / C++ Frameworks

- AMD Xilinx Vitis HLS (Vivado HLS)
- Catapult HLS (Siemens)
- Intel HLS Compiler

Python Frameworks

- hls4ml
- PyMTL3 / PyRTL
- Migen / LiteX

Rust Frameworks

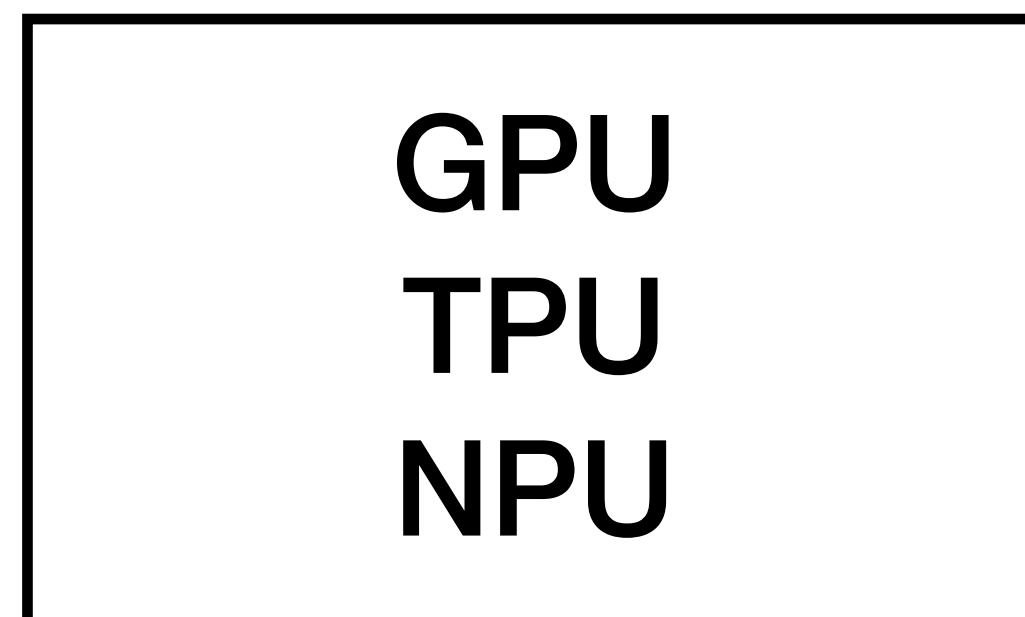
- Spade
- Veryl
- Rust-HDLs

Soft-Core CPUs & NPU

- Nios2 (Intel)
- Microblaze (Xilinx)
- RISC-V (Otvorený štandard)

Neurónové siete a FPGA

Trénovanie



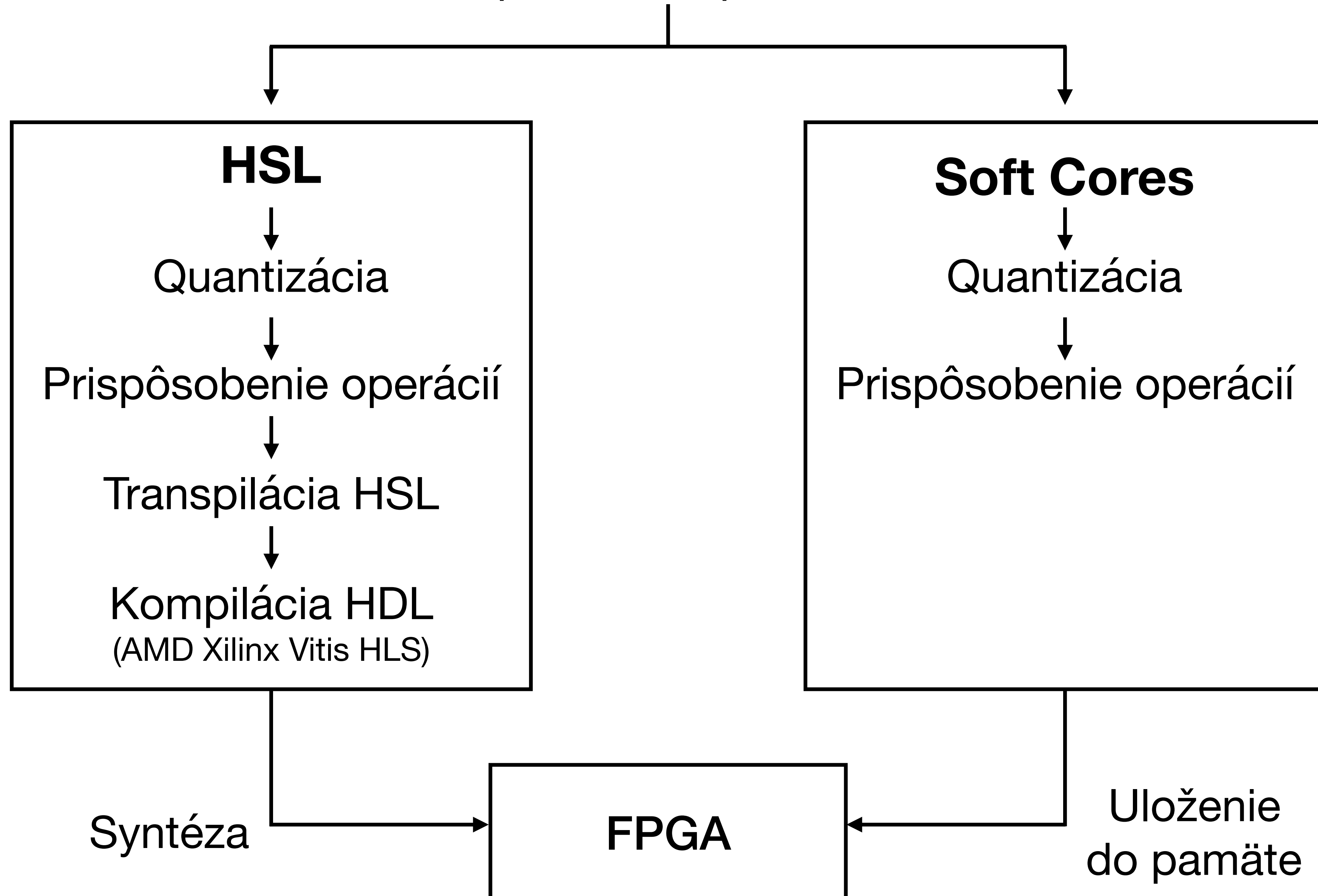
Prispôsobenie pre FPGA



Inference



Prispôsobenie pre FPGA



Triple Modular Redundancy